

John Gabriel Alcantara

Full-Stack Developer | Embedded Systems | IT Support

IT student and hands-on developer with experience spanning full-stack web development, embedded systems, IoT, hardware prototyping, and technical support. Completed a hands-on internship with MindBridge XR in Qatar Science & Technology Park, contributing across Engineering and IT through frontend, backend, and ESP32 firmware development. Experienced with sensor integration, hardware troubleshooting, APIs, databases, and real-time data systems. Also independently developed and improved a GenAI application for a startup founder. Strong practical problem-solving skills with experience working across hardware and software.

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Experience

Full-Stack & Embedded systems Intern

2026

MindBridge XR - Qatar Science and Technology Park(QSTP)

- Worked across the Engineering and IT departments, serving as a primary Arduino and embedded-systems resource.
- Contributed to frontend, backend, and firmware development for a sensor-based technology project.
- Developed and tested ESP32/ESP32-S3 firmware for collecting and transmitting sensor data.
- Integrated and tested 9-DoF motion sensors using I²C communication and breadboard-based hardware prototypes.
- Worked with real-time sensor-data workflows involving MQTT, backend services, and database storage.
- Contributed to web application development using React/Vite and backend services using Python/FastAPI.
- Worked with PostgreSQL and SQLAlchemy for application data management.
- Diagnosed and resolved hardware, firmware, software, sensor, and connectivity issues during development.
- Assisted with technical documentation and project handover.

Independent Full-Stack Developer

2026

Dreamveil - Independent Project

- Developed and improved a GenAI dream-storyboard web application targeting Windows, Android, and iOS.
- Upgraded the application's image-generation model to Go Banana 2.
- Implemented image-saving functionality for generated content.
- Diagnosed and fixed multiple bugs across the application.
- Worked independently from requirements provided directly by the founder/CEO of Sports Vector.
- Delivered completed improvements to an existing application while working across the full stack.

Freelance Computer Technician

2024 - Present

Independent

- Diagnosed, repaired, and upgraded 10+ computers and laptops for individual clients.
- Performed Windows installations, driver configuration, BIOS setup, hardware installation, and system troubleshooting.
- Conducted data recovery and assisted with performance optimization on damaged or malfunctioning systems.
- Troubleshoot hardware and software issues based on symptoms, testing, and system diagnostics.
- Provided technical guidance to users with varying levels of computer experience.

Technical Head

2023 - 2024

Youth for Christ Qatar

- Managed technical and audiovisual requirements for live events and organizational activities.
- Set up, operated, and troubleshooted computers, displays, audio equipment, and other technical systems.
- Coordinated technical requirements with event teams and responded to issues during live activities.

Simulated Patient

2025 - 2026

Qatar University

- Participated in medical education and clinical simulation activities requiring consistency, professionalism, and clear communication.
- Followed detailed scenarios and instructions while interacting with university students and faculty.
- Demonstrated reliability and adaptability in structured professional environments.

Education

Bachelor of Science in Information Technology

2023 - Present

University of Doha for Science and Technology

Qatar

Skills

Programming & Development

Python • C++ • JavaScript • HTML/CSS • Full-Stack Web Development • REST APIs • Application Debugging

Frontend Development

React • Vite • Web Application Development • Responsive UI Development • Real-Time Data Visualization

Backend & Databases

Python / FastAPI • SQLAlchemy • PostgreSQL • SQL • MQTT • API Development

Embedded Systems & IoT

Arduino • ESP32 / ESP32-S3 • Embedded C/C++ • Firmware Development • I²C • Sensor Integration • IMU / 9-DoF Sensors • MQTT / IoT Communication • Breadboard Prototyping

IT & Hardware

PC Assembly • Hardware Installation • Hardware Diagnostics • Hardware Troubleshooting • Windows Installation & Deployment • Driver Configuration • BIOS Configuration • Data Recovery • System Performance Optimization • End-User Technical Support

Tools & Software

Arduino IDE • VS Code • Git / GitHub • PostgreSQL • Microsoft Excel • CapCut

Projects

STINGRAY — Sensor-Based Rehabilitation System </> 2026
Full-Stack / Embedded Systems Project

Dreamveil — GenAI Dream Storyboard </> ↗ 2026
Independent Full-Stack Project | 2026

High School Capstone Project (Automatic Fire Suppression System for Cars) </> 2022
Lead Programmer / System Builder

Certifications

📄 **Certificate of Appreciation, AgriTeQ Exhibitor** - University of Doha for Science and Technology, 2026
Volunteered as an exhibitor for UDST's agricultural projects in AgriTeQ

📄 **4th Best Abstract, HS Capstone Project** - Philippine International School - Qatar, 2022
Awarded with the 4th best constructed abstract in PISQ's Capstone Project competition

📄 **3rd Overall HS Capstone Project** - Philippine International School - Qatar, 2022
placed 3rd overall in PISQ's Capstone Project competition.